Standard 2 Annuities

Question Bank

Question 1

Victoria wants to help her friend Irene compare compound interest options using a table that shows the future value interest factors of a compound interest investment of $1.

	Rate			
Period	*1%*	*2.5%*	*4%*	*5%*
5	1.0510	1.1314	1.2167	1.2763
10	1.1046		1.4802	1.6289
20	1.2202	1.6386	2.1911	2.6533

a) Victoria forgot to fill one of the entries in her table before giving it to Irene, what factor should go in the blank cell in the table?
b) Victoria is going to invest $4000 in an account that pays 5% p.a. compounding every 6 months. Irene is going to invest $4150 in an account that pays 4% p.a. compounding quarterly. Who will have more money in their account at the end of 5 years?

Question 2

Daphne hopes to have $30000 in a savings account in 5 years. The account pays 4% interest per annum, compounding quarterly. How much should she place into the account now to have the required amount at the end of 5 years?

Question 3

A table showing the future value interest factors of an annuity of $1 is shown below.

	Rate			
Period	*0.05%*	*0.1%*	*0.2%*	*0.4%*
135	139.6244	144.4596	154.8056	178.5399
180	188.2993	197.1097	216.4071	262.8712
270	288.9963	309.7877	357.5409	484.5857
390	430.5035	476.6930	589.8868	936.0091

Cynthia is planning to save regularly and is comparing two different saving schemes. Scheme A is to save $1200 monthly in an account that pays 2.4% per annum compounding monthly. Scheme B is to save $500 fortnightly in an account that pays 2.6% per annum compounding fortnightly. Which scheme would have a higher future value after 15 years?

Question 4

A table showing the future value interest factors of an annuity of \$1 is shown below.

	Rate			
Period	*0.001*	*0.002*	*0.003*	*0.004*
60	61.8047	63.6809	65.6316	67.6602
120	127.4292	135.4722	144.1857	153.6320
180	197.1097	216.4071	238.2067	262.8712
240	271.0967	307.6500	350.7400	401.6750

Agatha plans to save \$600000 by making monthly payments into an account that pays 3.6% interest per annum compounding monthly.
a) If she wants to save \$1710.67 each month, how many years will it take her to achieve her goal?
b) Agatha decides that she wants to save the amount in 15 years, how much will she have to save each month to make it happen?

Question 5

Below is a table showing the present value factor for an annuity paying \$1.

	Rate			
Period	*0.001*	*0.002*	*0.003*	*0.004*
60	58.2072	56.4866	54.8349	53.2489
120	113.0264	106.5918	100.6492	95.1560
180	164.6547	151.0364	138.9268	128.1370
240	213.2778	190.4600	170.9076	154.0933
300	259.0707	225.4297	197.6274	174.5210
360	302.1982	256.4488	219.9517	190.5977

a) Leona is to receive a pension payment of \$2400 per month from an account that pays 2.4% interest per annum compounded monthly for 20 years. What would the present value of the plan be if she were to receive a lump sum of money today?
b) Hannah takes a loan of \$60000 at an interest rate of 4.8% per annum compounded monthly. She is going to make monthly payments for 15 years, how much does she have to pay each month to pay off the loan?

Question 6

Rebecca wants to save some money for the future, so she puts $20000 into an account that pays 2.2% interest per annum compounding monthly. After 4 years, the bank increases the interest rate to 2.8% per annum and Rebecca decides to put an extra $5000 into the account for the next 5 years. How much money will she have in the account at the end of the 9 years?

Question 7

Below is a table showing the future value factor for an annuity of $1.

	Rate			
Period	*0.1%*	*0.2%*	*0.3%*	*0.4%*
180	197.1097	216.4071	238.2067	262.8712
216	240.9684	269.8354	303.2867	342.1367
252	286.4340	327.2482	375.7771	433.6529

Below is a table showing the present value factor for an annuity paying $1.

	Rate			
Period	*0.1%*	*0.2%*	*0.3%*	*0.4%*
90	86.0277	82.2898	78.7706	75.4556
120	113.0264	106.5918	100.6492	95.1560
150	139.2275	129.4799	120.6474	112.6328

Lydia wants to have some money saved for when her child turns 18. When her child is born, she saves $10000 into an account that pays 3.6% interest per annum compounding monthly. During this time she also saves $100 per month into the account.
a) When her child turns 18 how much money will be in the account?
b) Once her child is 18, the account will pay out a fixed amount every month for the next 10 years. How much will each payment be?

Question 8

Below is a table showing the present value factor for an annuity paying $1.

	Rate			
Period	*0.0025*	*0.003*	*0.0035*	*0.004*
60	55.6524	54.8349	54.0339	53.2489
120	103.5618	100.6492	97.8489	95.1560
180	144.8055	138.9268	133.3777	128.1370
240	180.3109	170.9076	162.1874	154.0933
300	210.8765	197.6274	185.5486	174.5210

Felicity takes out a loan with an interest rate of 4.2% per annum compounded monthly. The payment plan is to pay $1200 per month for the first 10 years and then $900 per month for the next 15.

a) What is the loan value after the first 10 years?

b) The original value of the loan would be the present value of value at 10 years, since it has been compounding, plus the value of the annuity being paid for the first 10 years. How much was the original value of the loan?

c) An alternative method for calculating the original loan amount might not make as much sense, but we can check that it gives the same answer. Split the loan payments into a plan where we pay $900 for 25 years, and also pay $300 for the first 10 years. The original loan should be the sum of the present value of these two annuities. How much is it using this method?

Answers

Question 1

a) Using the compound interest formula

$$(1 + 0.025)^{10} \approx 1.2801.$$

b) Victoria's investment will use the table value 1.2801 so her final value will be $1.2801 \times 4000 = \$5120.40$.
Irene's investment will use the table value 1.2202 so her final value will be $1.2202 \times 4150 = \$5063.82$.
Hence Victoria will have more in her account at the end of 5 years.

Question 2

Let her initial investment be the principle value P, the formula tells us

$$30000 = P\left(1 + \frac{4\%}{4}\right)^{5\times 4} = P \times 1.01^{20}$$

Hence her initial amount would be

$$P = 30000 \div 1.01^{20} = \$24586.33$$

Question 3

For Scheme A, we need to use the factor 216.4071 giving a future value of $216.4071 \times 1200 = \$259688.52$.
For Scheme B, use $476.6930 \times 500 = \$238246.50$.
So Scheme A would have a higher future value.

Question 4

a) Doing $600000 \div 1710.67 = 350.73876... \approx 350.7400$, and referencing the table indicates that this would need 240 periods. Since she is saving monthly 240 periods is 20 years.
b) 15 years is 180 monthly periods, and the table gives us a factor of 238.2067. $600000 \div 238.2067 = \$2518.82$ saved each month.

Question 5

a) The lump sum would be $2400 \times 190.4600 = \$457104$
b) $60000 \div 128.1370 = \$458.25$ per month.

Question 6

For the amount after first 4 years A_1,

$$A_1 = 20000\left(1 + \frac{0.022}{12}\right)^{4\times 12} = \$21838.00$$

Then she saves an extra \$5000, $\$21838 + \$5000 = \$26838$. So after the next 5 years

$$A = 26838\left(1 + \frac{0.028}{12}\right)^{5\times 12} = \$30866.01$$

Question 7

a) This will be a combination of a compound interest problem and annuity problem. For the interest on the \$10000,

$$A_c = 10000\left(1 + \frac{0.036}{12}\right)^{18\times 12} = \$19098.60.$$

Then for the annuity, we need to use the table value 303.2867,

$$A_a = 100 \times 303.2867 = 30328.67.$$

The total amount in the account at the end of 18 years will be

$$A = A_c + A_a = \$19098.60 + \$30328.67 = \$49427.27.$$

b) The annuity present value is \$49427.27, and since it will pay $\$x$ for 10 years we will use the factor 100.6492.

$$\begin{aligned} \$49427.27 &= x \times 100.6492 \\ \$49427.27 \div 100.6492 &= x \\ \$491.08 &= x. \end{aligned}$$

Question 8

a) Use the present value factor 133.3777,

$$A_{10} = \$900 \times 133.3777 = \$120039.93.$$

b) The present value of the amount left after 10 years A_{10} is given by the compound interest formula,

$$A_a = A_{10} \div 1.0035^{10\times 12} = \$120039.93 \div 1.0035^{10\times 12} = \$78929.71.$$

The present value of the annuity being paid for the first 10 years

$$A_b = \$1200 \times 97.8489 = \$117418.68.$$

The original value of the loan would be the sum of these present values,

$$\$78929.71 + \$117418.68 = \$196348.39.$$

c) The annuity of \$900 for 25 years would be

$$A_c = \$900 \times 185.5486 = \$166993.74.$$

The annuity of \$300 for 10 years would be

$$A_d = \$300 \times 97.8489 = \$29354.67.$$

The original present value of the loan should be the sum of the two,

$$\$166993.74 + \$29354.67 = \$196348.41.$$

We can see this method also gets about the same amount, but might be a bit more difficult to understand.